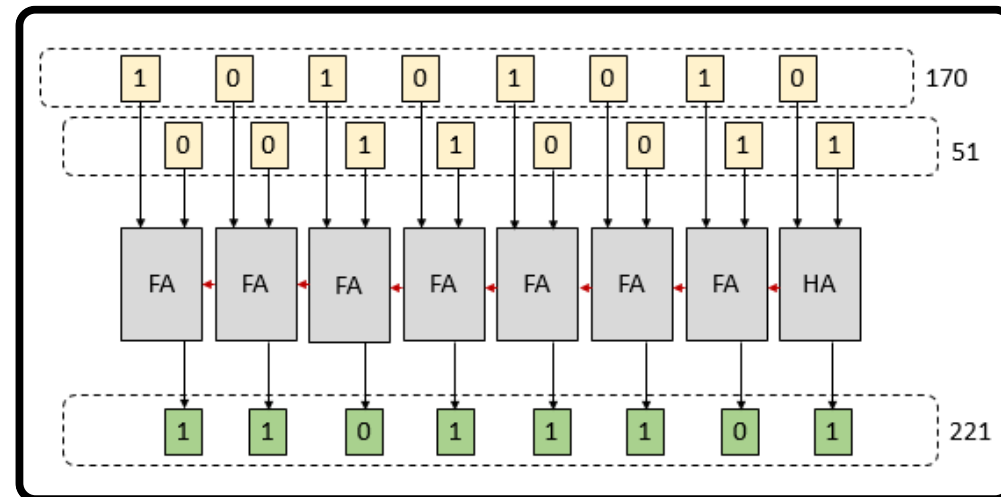


IEEE CONFERENCE PAPER PRESENTATION

DESIGN AND PERFORMANCE ANALYSIS OF 8-BIT CARRY SELECT ADDER USING HYBRID FULL ADDER

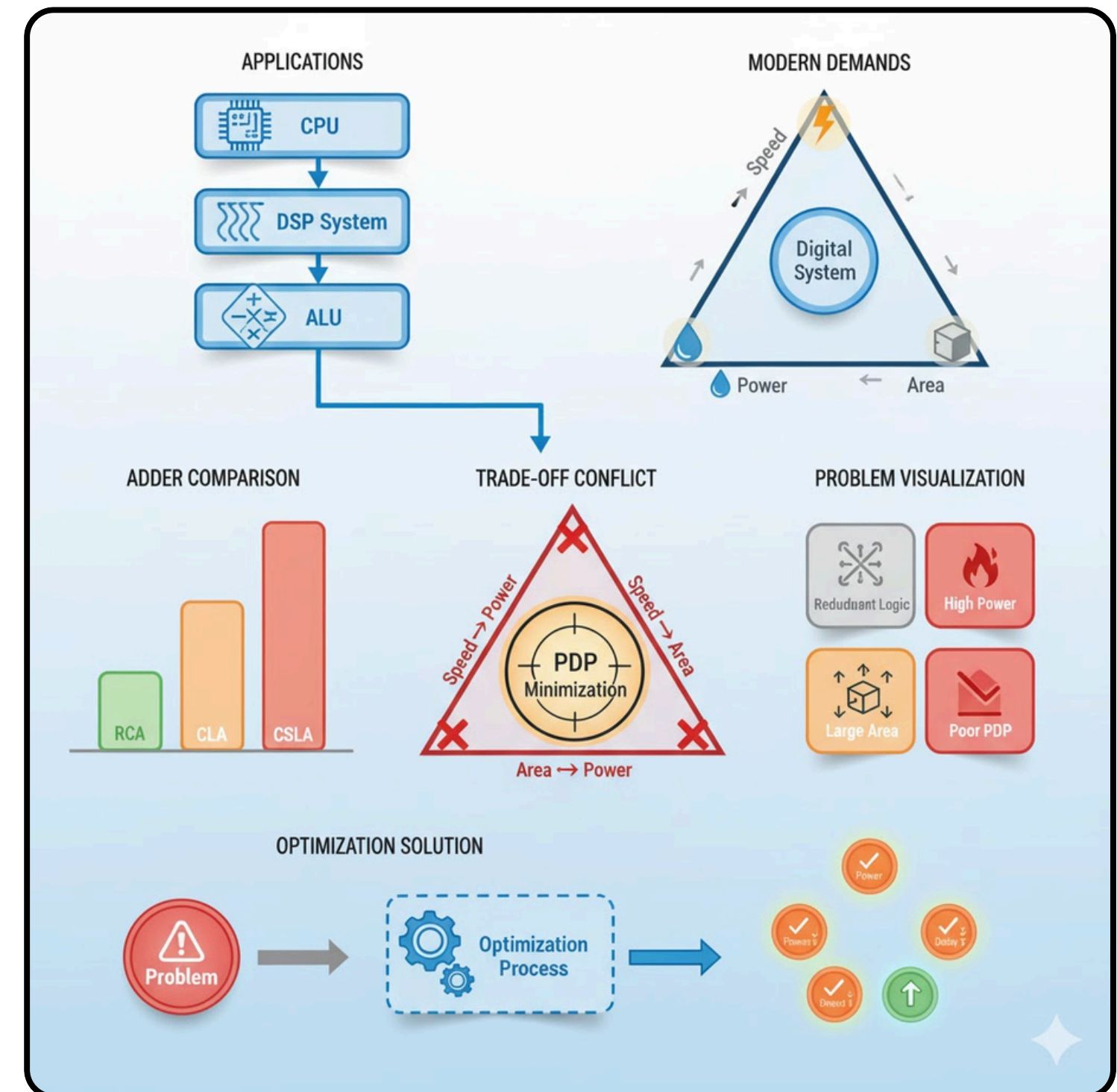
Authors

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PROBLEM STATEMENT

- Adders are fundamental building blocks in ALUs, DSP systems, and microprocessors
- Modern digital systems demand arithmetic circuits with **high speed**, **low power consumption**, and **minimal area**.
- Challenge:** Conventional Carry Select Adders (CSLAs) improve speed compared to Ripple Carry Adders (RCAs) and Carry Look-Ahead Adders (CLAs) but at the **expense of increased area and power dissipation due to redundant logic**.
- This performance trade-off becomes critical in VLSI designs where optimizing the Power–Delay Product (PDP) is essential.



LITERATURE SURVEY & GAP ANALYSIS

- **Classical Adder Architectures**

Adder	Technique	Pros	Cons	Ref
RCA	Cascaded full adders	✓ Simple, minimal area	✗ Slow ($O(n)$ delay)	[1][2]
CLA	Generate/Propagate logic	✓ Fast ($O(\log n)$)	✗ Complex, high area & wiring	[2][3]
Conventional CSLA	Duplicate adders for $C_{in}=0$ & $C_{in}=1$	✓ Faster than RCA ($O(\sqrt{n})$)	✗ Redundant logic → higher area & power	[3][4]

- **Optimized CSLA Designs**

Variant	Technique	Pros	Cons	Ref
BEC-based CSLA	Replace one RCA with Binary-to-Excess-1 Converter	✓ ~18% less area, ~26% lower power	✗ Still redundant logic, sub-optimal PDP	[4][5]
Low-Power CSLA / Logic Restructuring	Logic sharing, gating, optimized multiplexers	✓ Better PDP than standard CSLA	✗ Higher design complexity	[5]

LITERATURE SURVEY & GAP ANALYSIS

• Full Adder Cell Designs

FA / Adder Type	Transistors	Pros	Cons / Limitations	Effect in CSLA	Ref
Standard CMOS FA	28	✓ Robust, full swing	✗ High transistor count → larger area & power	Baseline CSLA → higher PDP	[6]
Hybrid Full Adder (HFA)	20-22	✓ Low transistor count, fast, low	✗ Slightly more complex design	Reduces PDP significantly in CSLA	[6] [7]
NMOS/PMOS-only FA / PTL	10-16	✓ Very low transistor count	✗ Voltage degradation, extra buffers needed	Not practical for 8-bit CSLA	[6]

GAP ANALYSIS

- Classical adders (RCA, CLA) either suffer from **slow speed or high complexity/area**.
- Conventional CSLA reduces carry delay but **duplicates logic, increasing area and power**.
- Optimizations like BEC-CSLA or low-power logic improve some metrics, but often ignore transistor-level efficiency.
- Hybrid Full Adders (HFA) optimize power and transistor count, but are rarely applied to full CSLA architectures.
- **Key Gap: Most designs optimize either the architecture or the full adder cells, not both → overall PDP remains sub-optimal.**

METHODOLOGY

OUR SOLUTION

Implement 8-bit CSLA using Hybrid Full Adders

Hybrid
Architecture
PTL + GDI

- Technology:
GPDK 180nm
- Each HFA uses
22 MOSFETs

Boolean Logic
Implementation

$$\text{SUM} = A \oplus B \oplus C_{\text{in}}$$
$$\text{Cout} = AB + C_{\text{in}}(A \oplus B)$$

Optimization

- Cell-Level & Architecture-Level Efficiency

Reducing:

- Transistor Count
- Power Consumption
- Area & Delay

Maximizing Speed

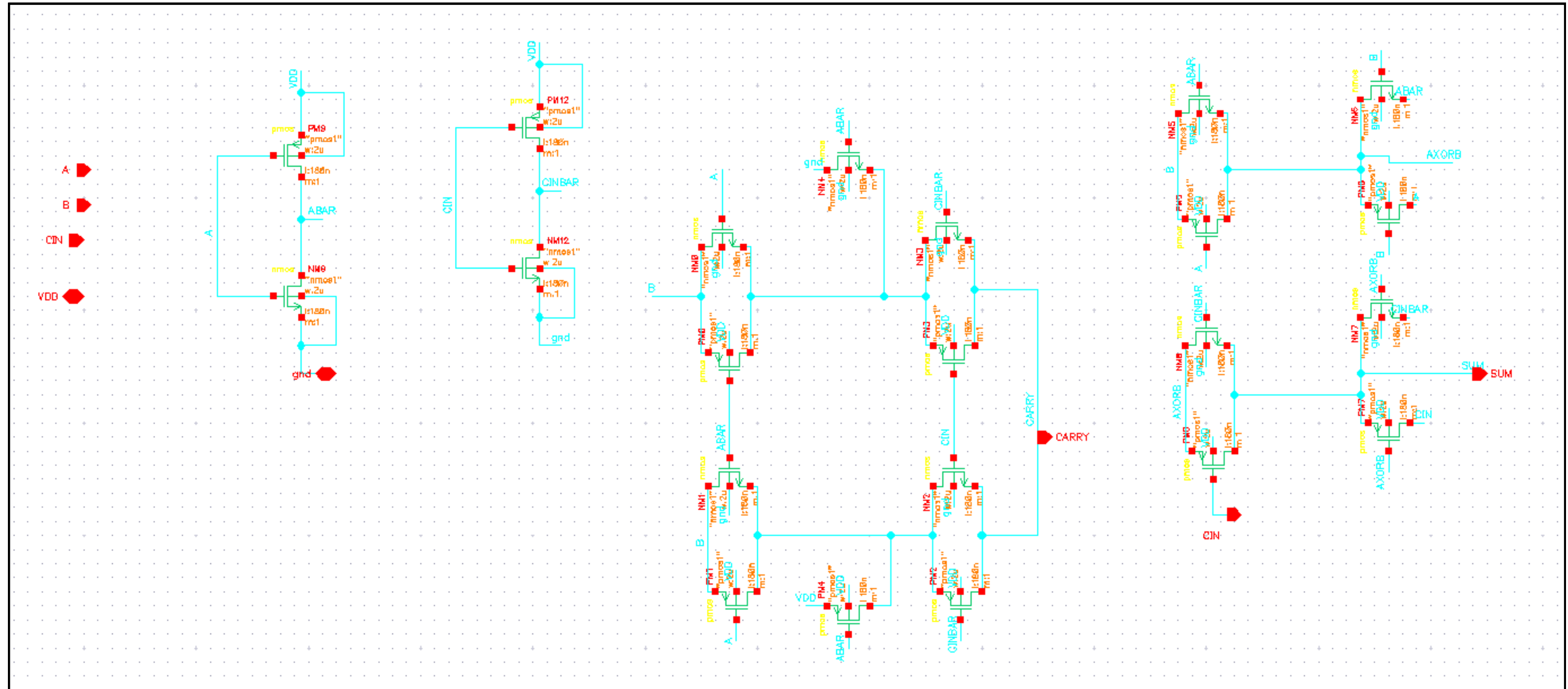
INPUT
 $A[7:0] + B[7:0] + C_{\text{in}}$

8 BIT CSLA
USING HFA

OUTPUT
 $\text{SUM}[7:0] + \text{Cout}$

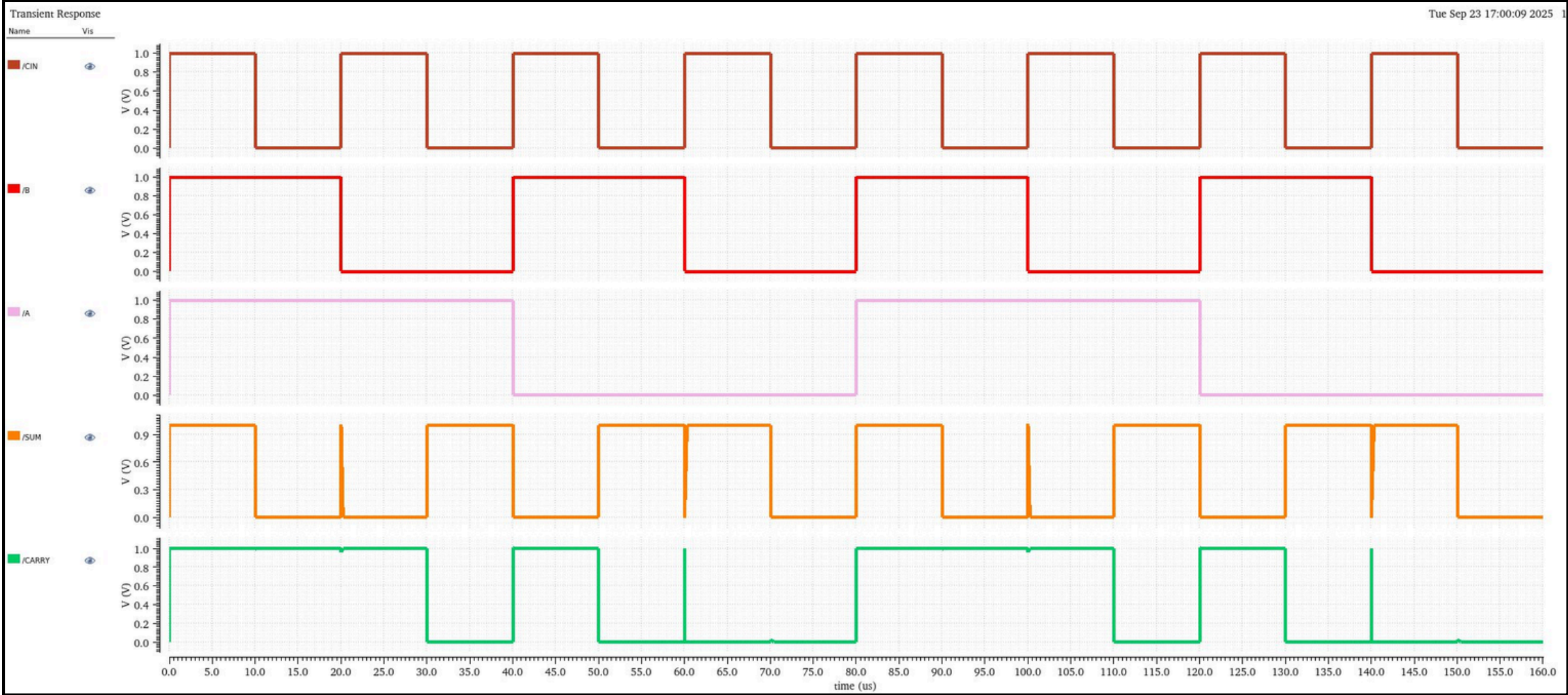
METHODOLOGY

DESIGN OF 1 BIT HYBRID FULL ADDER



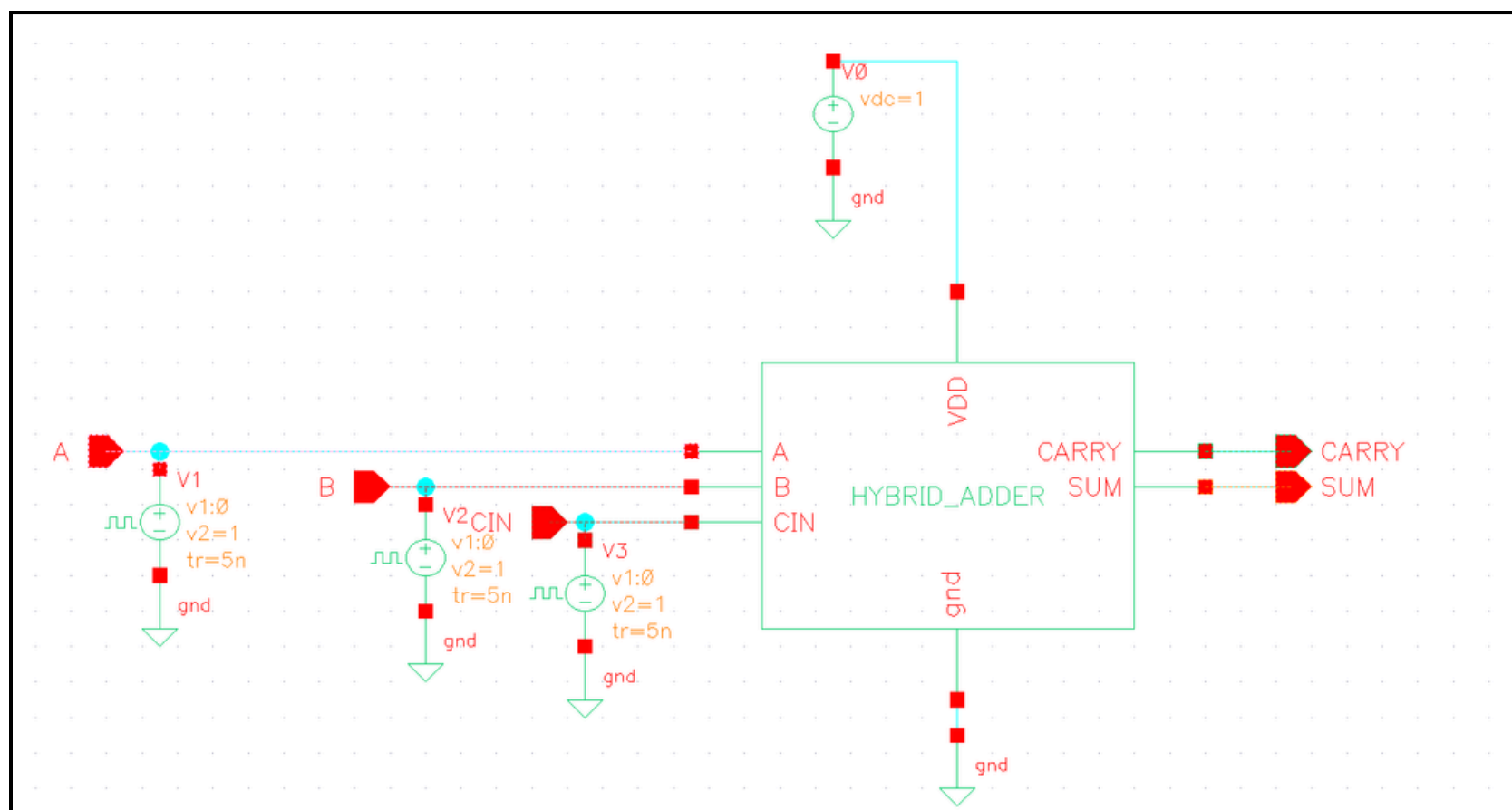
METHODOLOGY

INPUT & OUTPUT WAVEFORMS OF 1 BIT HYBRID FULL ADDER

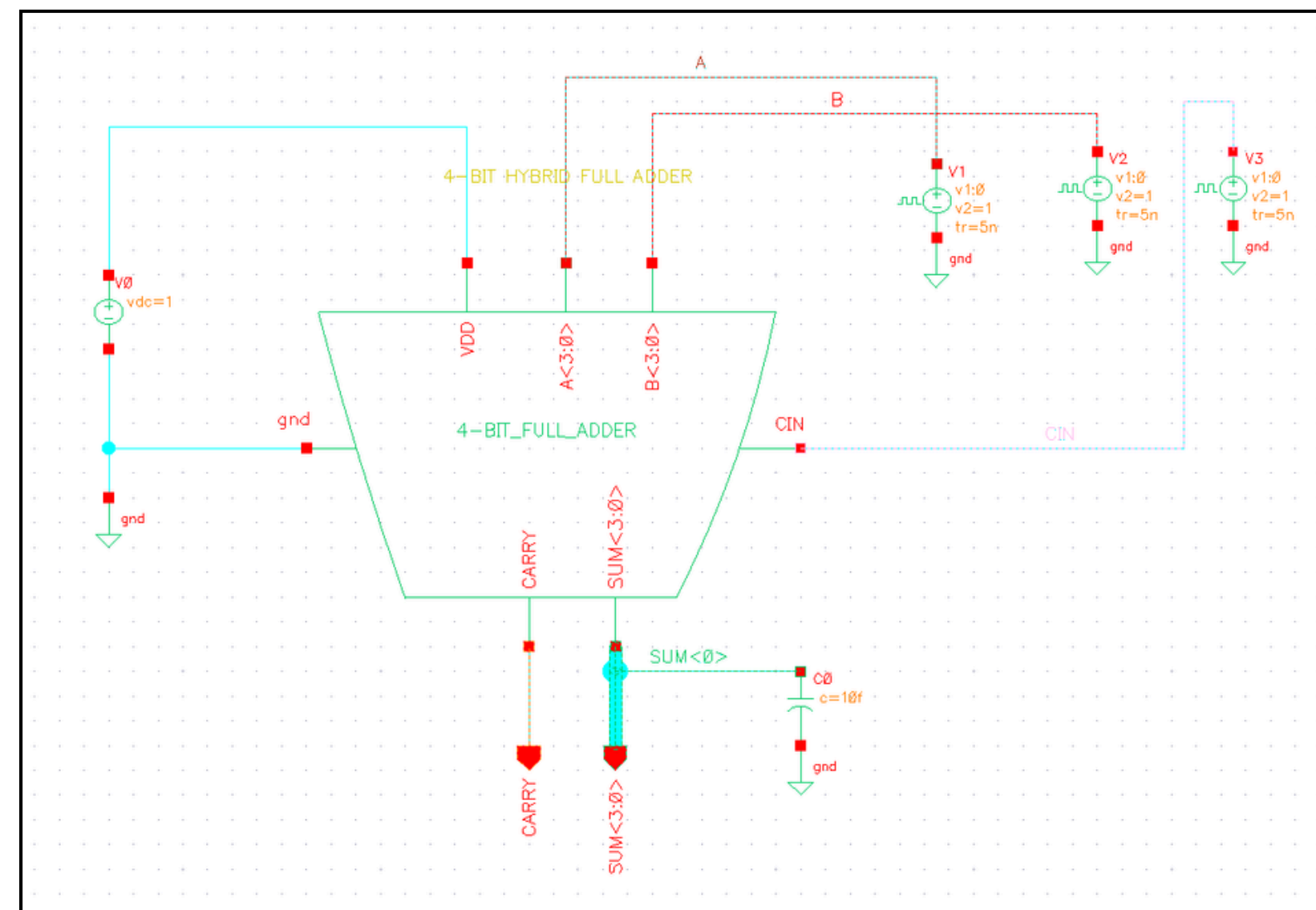


Input			Output	
A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

METHODOLOGY

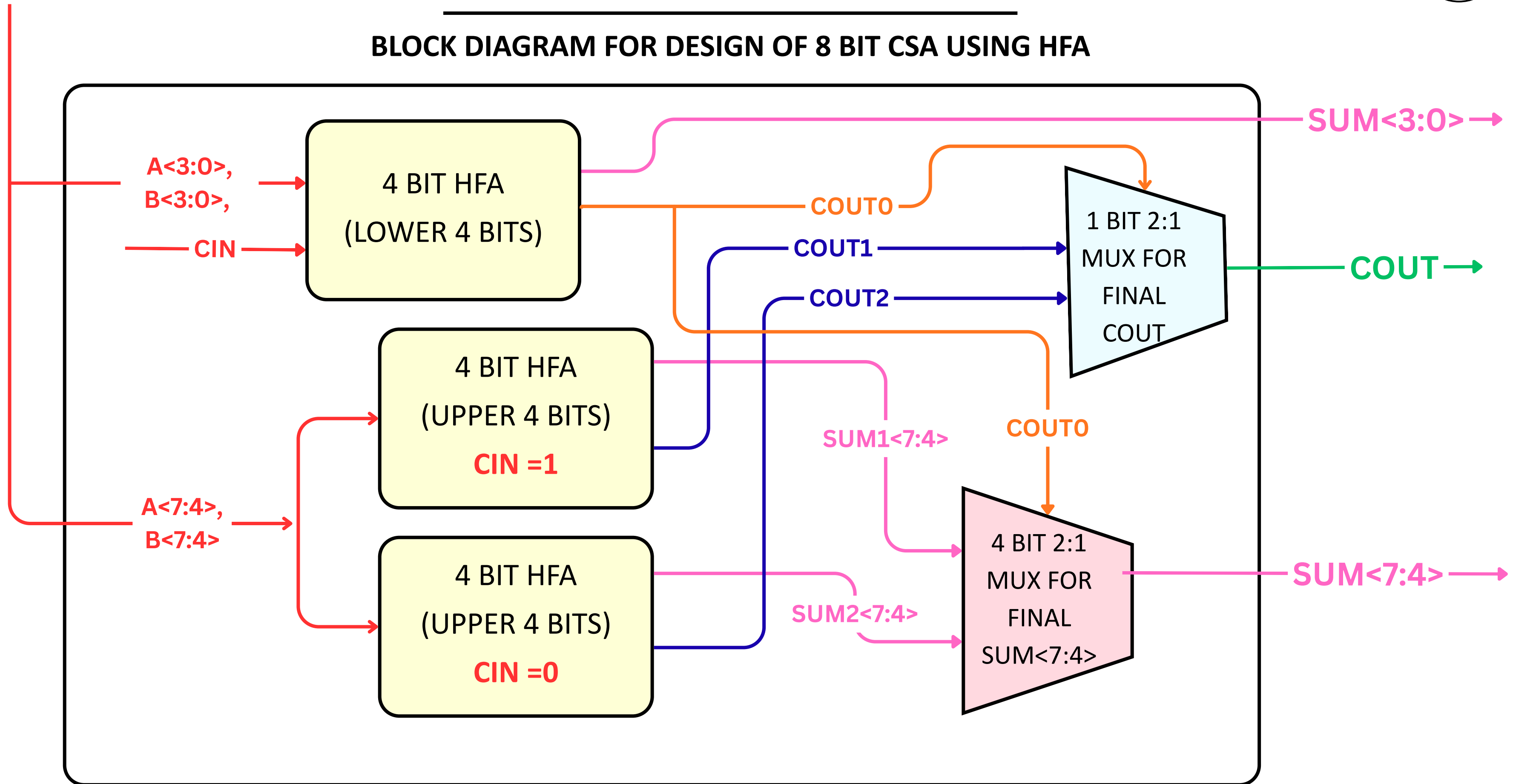


SYMBOL OF 1 BIT HYBRID FULL ADDER



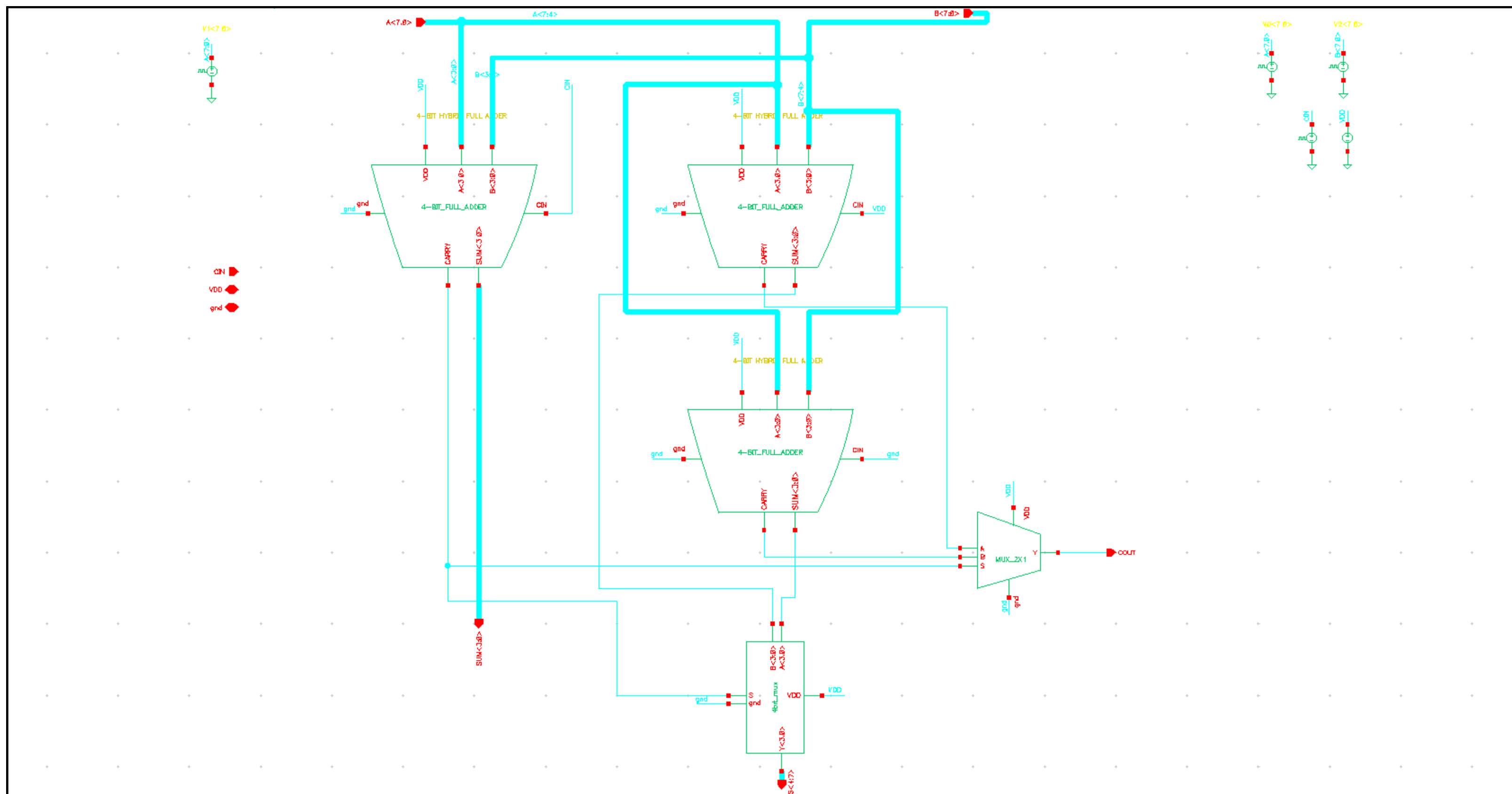
SYMBOL OF 4 BIT HYBRID FULL ADDER

BLOCK DIAGRAM FOR DESIGN OF 8 BIT CSA USING HFA



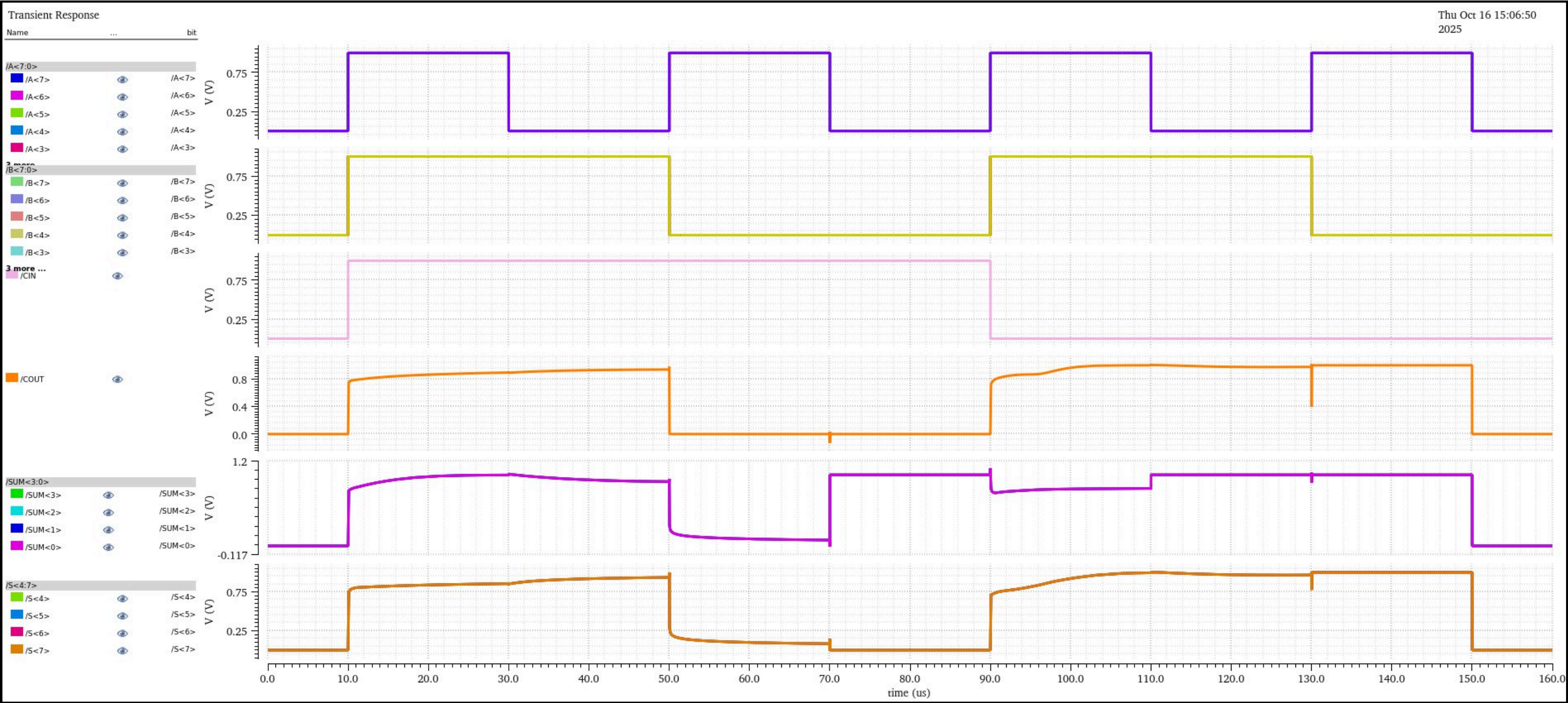
METHODOLOGY

SCHEMATIC DIAGRAM FOR 8 BIT CSA USING HFA

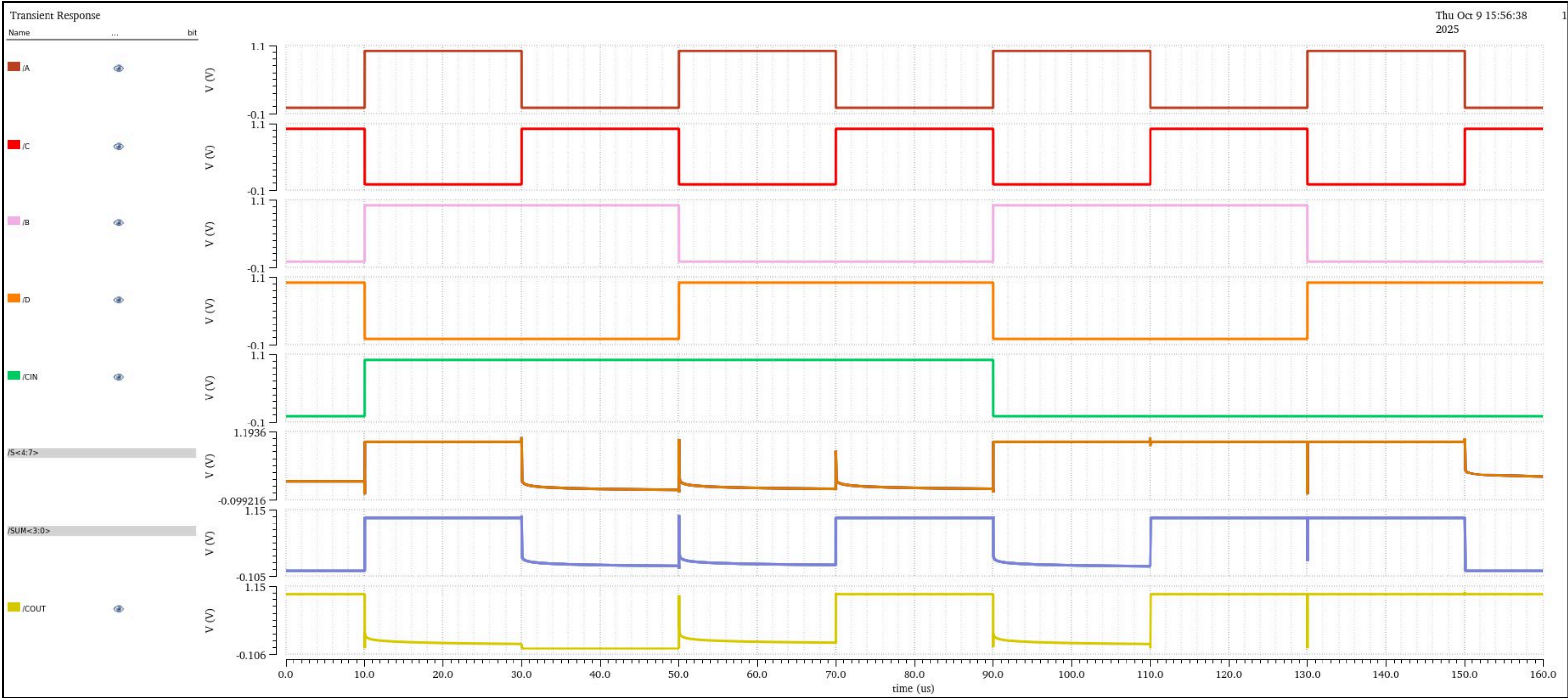


RESULTS

INPUT & OUTPUT WAVEFORMS FOR 8 BIT CSA USING HFA



INPUT & OUTPUT WAVEFORMS FOR 8 BIT CSA USING HFA



DISCUSSION OF RESULTS

- The proposed 8-bit CSLA using Hybrid Full Adder (HFA) works **correctly for all input combinations** as verified through simulation waveforms.
- Using HFA **reduced the total transistor count**, which directly decreased area and switching activity.
- The 8-bit HFA-CSLA successfully demonstrates functional operation at 1.0V with **1.8 μ W power and 6.9 ns delay**.
- **$PDP = Power \times Delay = 1.8 \mu W \times 6.9 ns = 12.42 fJ$**
- The design validates feasibility of hybrid adder cells in **near-threshold operation**.
- Overall, the results prove that optimizing both the cell level (HFA) and architecture level (CSLA) provides higher speed and lower power on the same design.

CONCLUSION

- The proposed HFA-based CSLA overcomes the drawbacks of a conventional CSLA.
- It reduces power, delay, and transistor count, improving overall performance.
- The PDP is significantly improved, making the design more efficient for VLSI systems.
- The adder is suitable for high-speed and low-power applications like ALUs, DSP processors, and embedded systems.

FUTURE SCOPE

- Can be scaled to **16-bit, 32-bit, or 64-bit** adders for larger processors.
- Can be implemented in advanced nanometer CMOS technologies (**90nm, 45nm, etc.**) for even better results.
- Can be combined with **low-power techniques** like clock gating and multi-Vth optimization.
- Possible use in IoT devices, AI accelerators, DSP blocks, and portable electronics where power and speed are critical.
- Can be extended to approximate or hybrid adders for low-power AI and multimedia applications.

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Literature Survey & Gap Analysis

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2. GeeksforGeeks, "Carry Look-Ahead Adder," 2020.
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5. IJAERS, "Low Power and Area Efficient Carry Select Adder," 2016.
6. M. J. Smith, "Full Adder Transistor-Level Designs," ScienceDirect, 2021.
7. P. K. Singh, "Hybrid Full Adder Design for Low Power Applications," IET Research, 2023.

Methodology

https://youtu.be/eas-fPGSMew?si=2_4sdcSGYDvNk32C